

Similarity Index of 1D Cellular Automata

Graph approach

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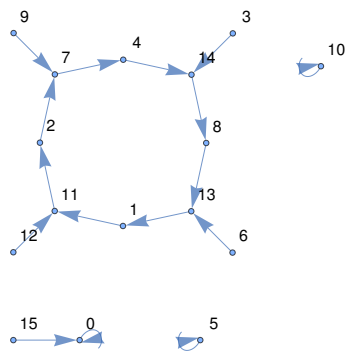
Spectral Moments

Instead of comparing the full spectrum, we compare a fixed number of spectral moments:

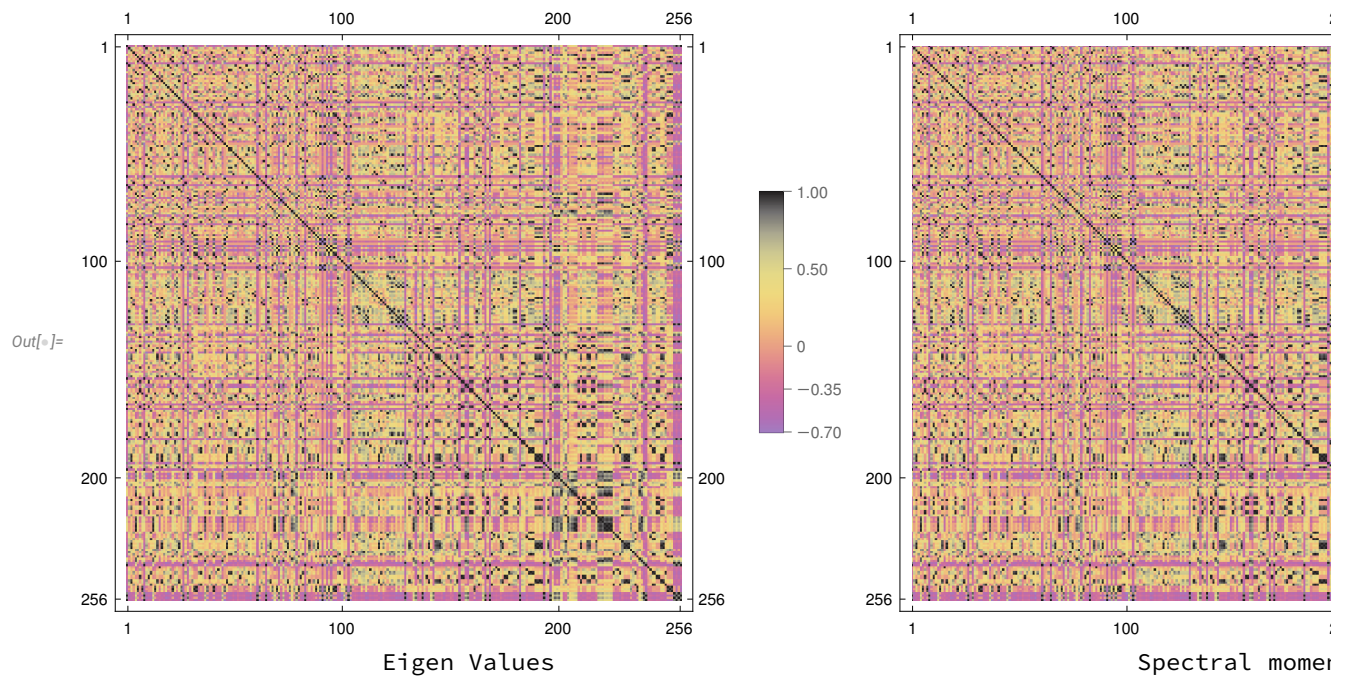
Run the rule j steps forwards on every possible configuration and define the fraction m_j of them that landed back exactly where they started.

The new feature vector is:

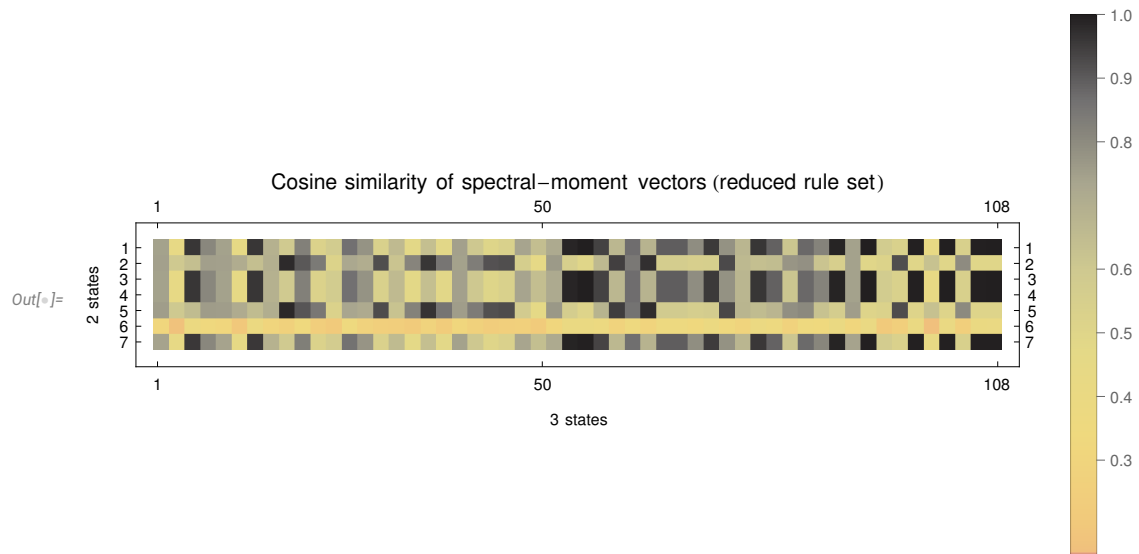
$$(m_1, m_2, m_3, \dots, m_K)$$

Out[$\#$]=		$\langle m_1 \rightarrow 3, m_2 \rightarrow 3, m_3 \rightarrow 3, m_4 \rightarrow 3, \\ m_5 \rightarrow 3, m_6 \rightarrow 3, m_7 \rightarrow 3, m_8 \rightarrow 11 \rangle$
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Spectral moments similarity



Distance between rules in different spaces



Thanks